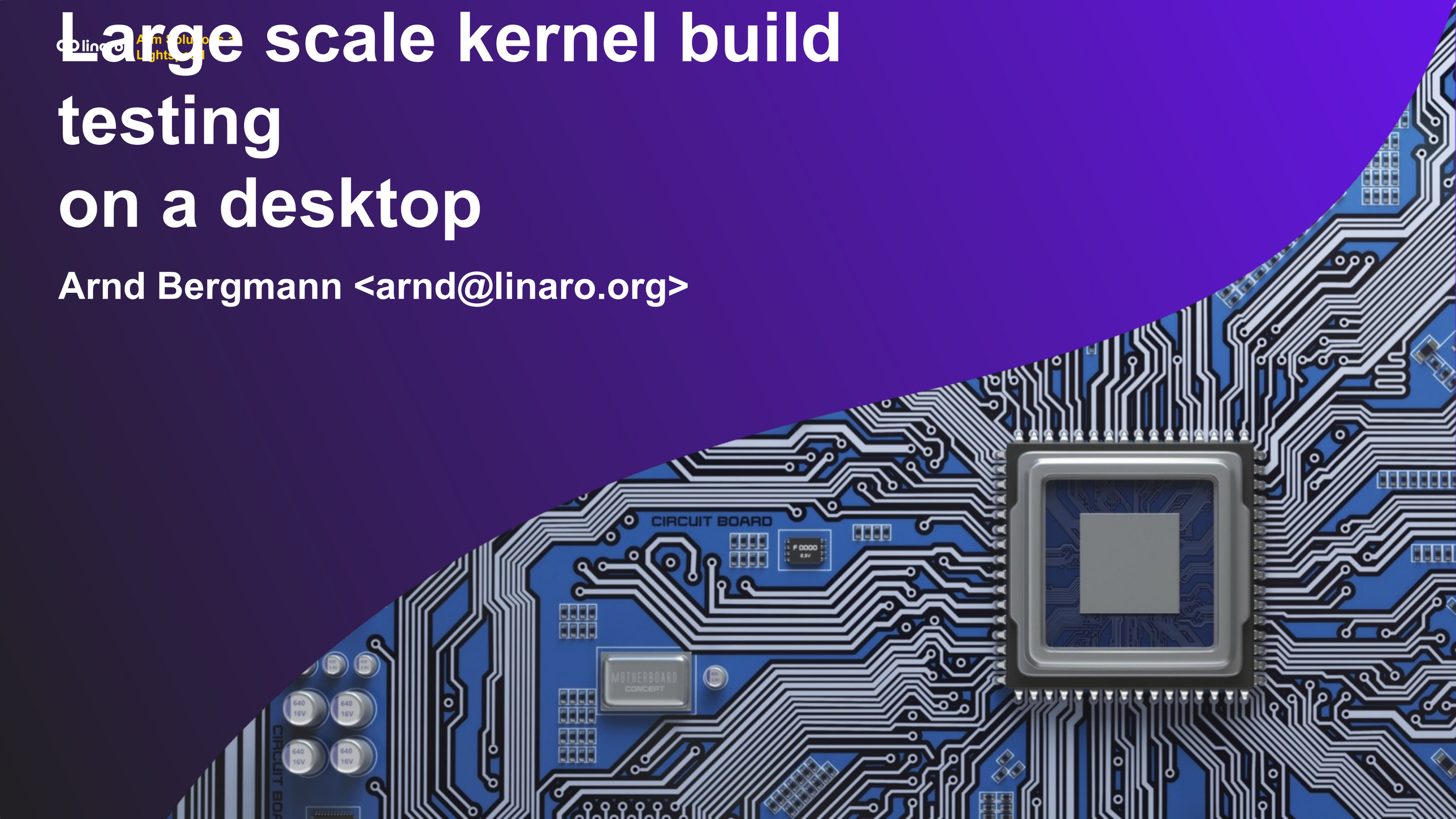
Large scale kernel build testing on a desktop

Arnd Bergmann <arnd@linaro.org>

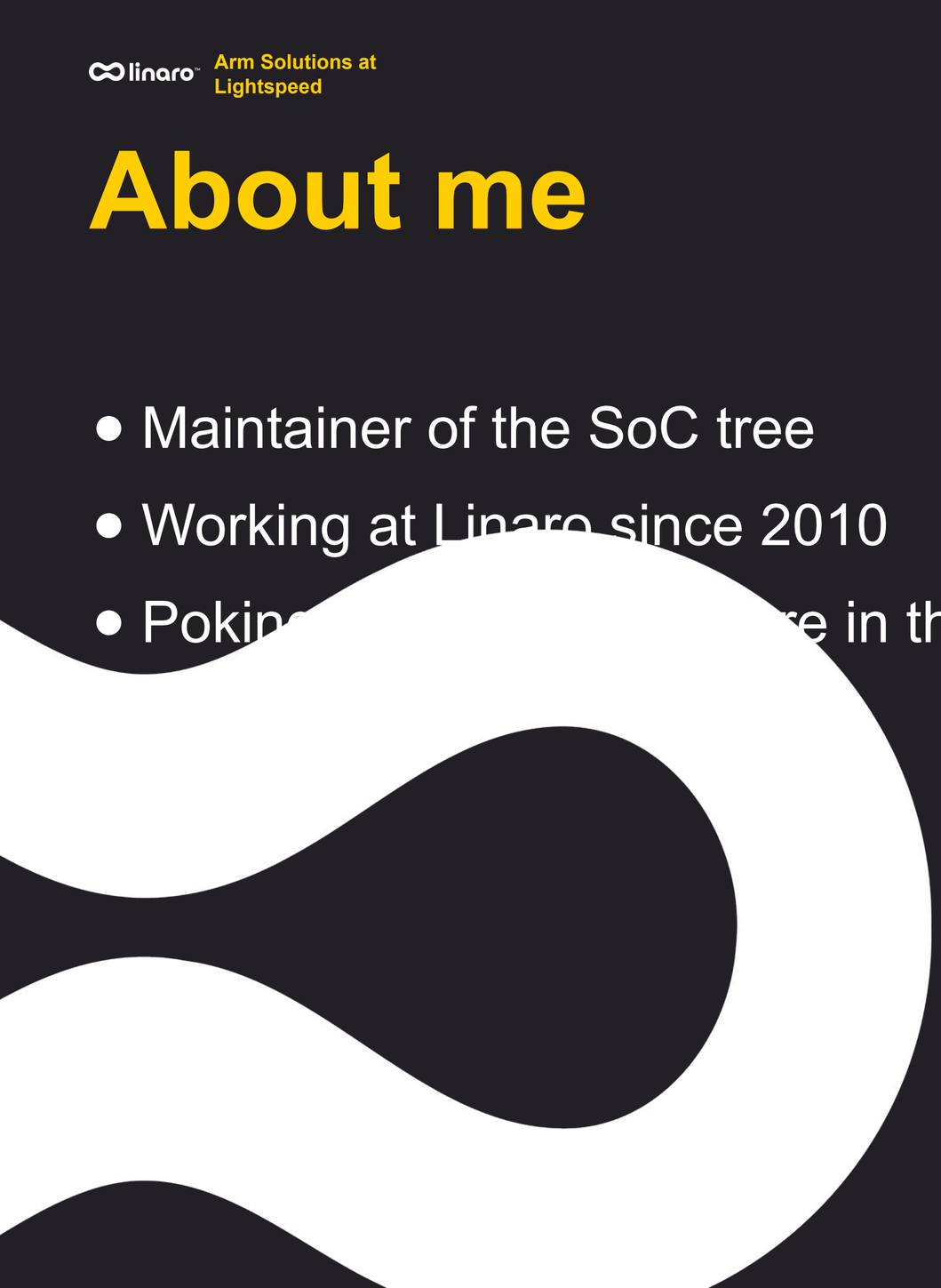




Introduction

About me

- Maintainer of the SoC tree
- Working at Linaro since 2010
- Pokin[illegible]re in the kernel



About me

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  you have

Motivation

Why create your own CI system?

Alternatives

- tuxsuite
- kernel
- …

Advantages of my approach

- No remote login
- Testing the SoC tree while merging
- Testing API Changes
- Faster turnaround
- Energy efficiency



Implementation

How to keep the expensive workstation occupied

Hardware

2024-2026 Apple Mac Studio M2 Ultra, 24 cores, 64GB LPDDR5

2022-2024 Apple Mac Studio M1 Ultra, 20 cores, 64GB LPDDR5

2017-20 per 1950

ECC DDR4

Dual AMD C 6344

24 cores, 128 C DDR3



Makefiles and scripts

- One top-l akefile
  - Inter akefile
  - ur el build jobs
  - One tree per job
  - Share e tree

Loop per architecture

- arm, arm64, x86, s390
- randconfig
- 300 builds in a row
- Filter out failed builds

Rebuild failed configs

- Retry in parallel
- fix/commit/submit

Workflow

- Keep machine busy testing linux-next
- Build with -Werror
- Fix as
- backlog to late -next
- Merge in my own proje

Example session

```
arnd@studio:~/arm-soc$ nice -n 19 make -skj20 randconfigs
0x40F225B2 Do 17. Sep 03:45:59 CEST 2026 success
0xF0498C20 Do 17. Sep 03:46:49 CEST 2026 success
0x765EB222 Do 17. Sep 03:47:49 CEST 2026 success
0x7CAC9420 Do 17. Sep 03:50:12 CEST 2026 success
0x5E2D2694 Do 17. Sep 03:54:12 CEST 2026 success
0xB8241975 Do 17. Sep 03:55:55 CEST 2026 success
0x9E26C30C Do 17. Sep 03:56:03 CEST 2026 success
0xA128D170 Do 17. Sep 03:56:09 CEST 2026 failed
/home/arnd/cross/arm64/gcc-16.2.0-nolibc/x86_64-linux/bin/x86_64-linux-ld: drivers/net/ethernet/sfc/efx_cxl.o: in function `efx_cxl_init':
efx_cxl.c:(.text+0xfc): undefined reference to `devm_cxl_probe_mem'
0xEE886ADF Do 17. Sep 04:00:43 CEST 2026 success
0xC9C8A56E Do 17. Sep 04:02:26 CEST 2026 success
0x17A6D7E Do 17. Sep 04:02:28 CEST 2026 failed
In function 'get_tier_idx',
    inlined from 'isolate_folios' at /home/arnd/arm-soc/mm/vmscan.c:4854:14:
/home/arnd/arm-soc/mm/vmscan.c:4812:9: error: 'sp.refaulted' is used uninitialized [-Werror=uninitialized]
 4812 |         read_ctrl_pos(lruvec, type, LRU_TIER_MIN, LRU_TIER_MIN, 2, &sp);
      |         ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
/home/arnd/arm-soc/mm/vmscan.c: In function 'isolate_folios':
/home/arnd/arm-soc/mm/vmscan.c:4824:25: note: 'sp.refaulted' was declared here
```

Example session

```
0xB03A7A69 Do 17. Sep 08:05:15 CEST 2026 success
0xD6291D20 Do 17. Sep 08:05:27 CEST 2026 success
0x4C445474 Do 17. Sep 08:13:19 CEST 2026 success
0xEBDD6A3E Do 17. Sep 08:13:20 CEST 2026 success
0xDB04E256 Do 17. Sep 08:13:24 CEST 2026 success
^Z
[1]+  Stopped                 make -skj20 randconfigs
arnd@studio:~/arm-soc$ make newrands ; make -skj20 buildall GCCVER=15.3.0 ; fg
'rand86/0xA128D170-config' -> 'arch/x86/configs/0xA128D170_defconfig'
'rand64/0x7D1B5E40-config' -> 'arch/arm64/configs/0x7D1B5E40_defconfig'
'rand64/0xB37A517-config' -> 'arch/arm64/configs/0xB37A517_defconfig'
'rand390/0x34351E5B-config' -> 'arch/s390/configs/0x34351E5B_defconfig'
'rand390/0x6B649E3C-config' -> 'arch/s390/configs/0x6B649E3C_defconfig'
'rand390/0x9548D07-config' -> 'arch/s390/configs/0x9548D07_defconfig'
'rand390/0xC7A824AE-config' -> 'arch/s390/configs/0xC7A824AE_defconfig'
Thu, 17 Sep 2026 08:14:47 +0200 build/arm/0xC63E0B9E_defconfig
Thu, 17 Sep 2026 08:14:47 +0200 build/x86/0x3E604DA0_defconfig
Thu, 17 Sep 2026 08:14:47 +0200 build/x86/0xA128D170_defconfig
Thu, 17 Sep 2026 08:14:48 +0200 build/x86/0xEE534190_defconfig
Thu, 17 Sep 2026 08:14:48 +0200 build/arm64/0x653C516D_defconfig
Thu, 17 Sep 2026 08:14:48 +0200 build/arm64/0x7D1B5E40_defconfig
```



Example session

```
arnd@studio:~/arm-soc$ $ make newrands ; make -skj20 buildall GCCVER=16.2.0 ; fg
Thu, 17 Sep 2026 16:24:39 +0200 build/arm/0xC63E0B9E_defconfig
Thu, 17 Sep 2026 16:24:39 +0200 build/x86/0x3E604DA0_defconfig
Thu, 17 Sep 2026 16:24:40 +0200 build/x86/0xA128D170_defconfig
Thu, 17 Sep 2026 16:24:40 +0200 build/x86/0xC97B6E84_defconfig
Thu, 17 Sep 2026 16:24:40 +0200 build/x86/0xEE534190_defconfig
Thu, 17 Sep 2026 16:24:40 +0200 build/arm64/0x653C516D_defconfig
Thu, 17 Sep 2026 16:24:40 +0200 build/arm64/0xB37A517_defconfig
Thu, 17 Sep 2026 16:24:40 +0200 build/arm64/0xC45640A8_defconfig
Thu, 17 Sep 2026 16:24:40 +0200 build/s390/0x34351E5B_defconfig
Thu, 17 Sep 2026 16:24:40 +0200 build/s390/0x6B649E3C_defconfig
Thu, 17 Sep 2026 16:24:40 +0200 build/s390/0x9548D07_defconfig
Thu, 17 Sep 2026 16:53:33 +0200 build/arm64/0xC45640A8_defconfig success
Thu, 17 Sep 2026 16:54:05 +0200 build/arm/0xC63E0B9E_defconfig fail
Thu, 17 Sep 2026 16:57:48 +0200 build/s390/0x9548D07_defconfig success
Thu, 17 Sep 2026 16:58:15 +0200 build/x86/0xEE534190_defconfig success
Thu, 17 Sep 2026 16:58:25 +0200 build/x86/0x3E604DA0_defconfig success
Thu, 17 Sep 2026 16:58:38 +0200 build/x86/0xC97B6E84_defconfig success
Thu, 17 Sep 2026 16:59:26 +0200 build/arm64/0x653C516D_defconfig success
Thu, 17 Sep 2026 16:59:31 +0200 build/x86/0xA128D170_defconfig fail
Thu, 17 Sep 2026 16:59:32 +0200 build/s390/0x34351E5B_defconfig fail
```



Toolchain

gcc

- all targets except hexagon
- all su
- ally for arm64/x 64
- uploaded to
  - //mirrors.edge.kernel.org/p osstool/

Clang

- built by Nathan Chancellor
- targets: arm64, arm, hexagon, loongarch, mips, powerpc, riscv, s390, x86
- uploaded to
  - https://mirrors.edge.kernel.org/pub/tools/llvm/

Configuration hacks

Fragment to avoid failed builds

$ cat kernel/configs/random.config

maximize search space, disable options not worth testing

CONFIG_COMPILE_TEST=y

avoid long build times

PROFILE_ALL_BRANCHES is not set

reduce compile-time dependencies

CONFIG_STANDALONE=y

CONFIG_PREVENT_FIRMWARE_BUILD=y



Backlog tree

arnd@studio:~/arm-soc$ git log --oneline
0a1affa74ef2 (HEAD -> randconfig-7.3-next) rtc: renesas: avouud -Wtautological-constant-out-of-range-compare warning
7e0d683ced55 [SUBMITTED 20260917] usb: gadget: pxa25x: prevent ixp4xx + pxa25x combination
4517c5008ca3 [SUBMITTED 20260917] sfc: fix CONFIG_CXL_MEM dependency
41ae3f9b5c68 [SUBMITTED 20260917] f2fs: quota: fix unused-label warning
2df8a5b2ff63 wait: avoid -Wconstant-logical-operand warning
9e6856c86c1a [SUBMITTED 20260917] [v2] ASoC: rt766: fix HID dependency for SND_SOC_SDCA_HID
9c6e65ae61e1 fixup vmscan
893426e48fa3 [SUBMITTED 20260917] nvme: work around all -Wformat-security warnings
00288e0405a3 fixup! treewide: address -Wformat-security warnings
2bfa4997a200 fixup! x86: kvm: drop 32-bit host support
5fa2a8a4d91c [SUBMITTED 20260916] mm/vmscan: avoid false-positive -Wuninitialized warning, again
d16ce85991bd [SUBMITTED 20260916] firmware: imx: fix build regression
1cb6bf9c09cd [SUBMITTED 20260916] nvme: work around -Wformat-security warning
cb0052abecbf [SUBMITTED 20260916] usb: fotg210: fix USB_EHCI_HCD dependency
958c0814889f [SUBMITTED 20260916] gpiolib: remove linux/gpio.h
015d784ac1ca [SUBMITTED 20260916] gpiolib: turn off legacy interface by default
40ba4190040e [SUBMITTED 20260916] riscv: limit sifive errata to CONFIG_64BIT
11326674e1de [SUBMITTED 20260916] media: i2c: it6625: select CONFIG_HDMI
9271e8a4e9cc fixup! arm64: make swiotlb optional
f4d03eeabc7d [SUBMITTED 20260916] net: dsa: motorcomm: fix unused-function warning



Bugs found so far



Bugs found so far

- Thousands of bugs fixed upstream
- Every subsystem touched
- More f lse

om/blog/kernel-bugs-part2

re patches fo o other work or dropped

Kconfig dependency errors

- Most common bug in randconfig
- Often not found in allmodconfig builds
  - missin
  - tri
  - et dependency
  - ency loop
- Nobody can do this cor
  - ope: https://git /julianbraha/kconfirm

Stack frame warnings

- Unsolved problem
- Hard limit of kernel thread stacks, usually 16KiB
- Per fu crude indicator
  - d to
  - neuristic mis
  - configurations
- Testing limit:
  - 4b + 256b CONFIG + 256b CONFIG_KASAN + 256b KMSAN

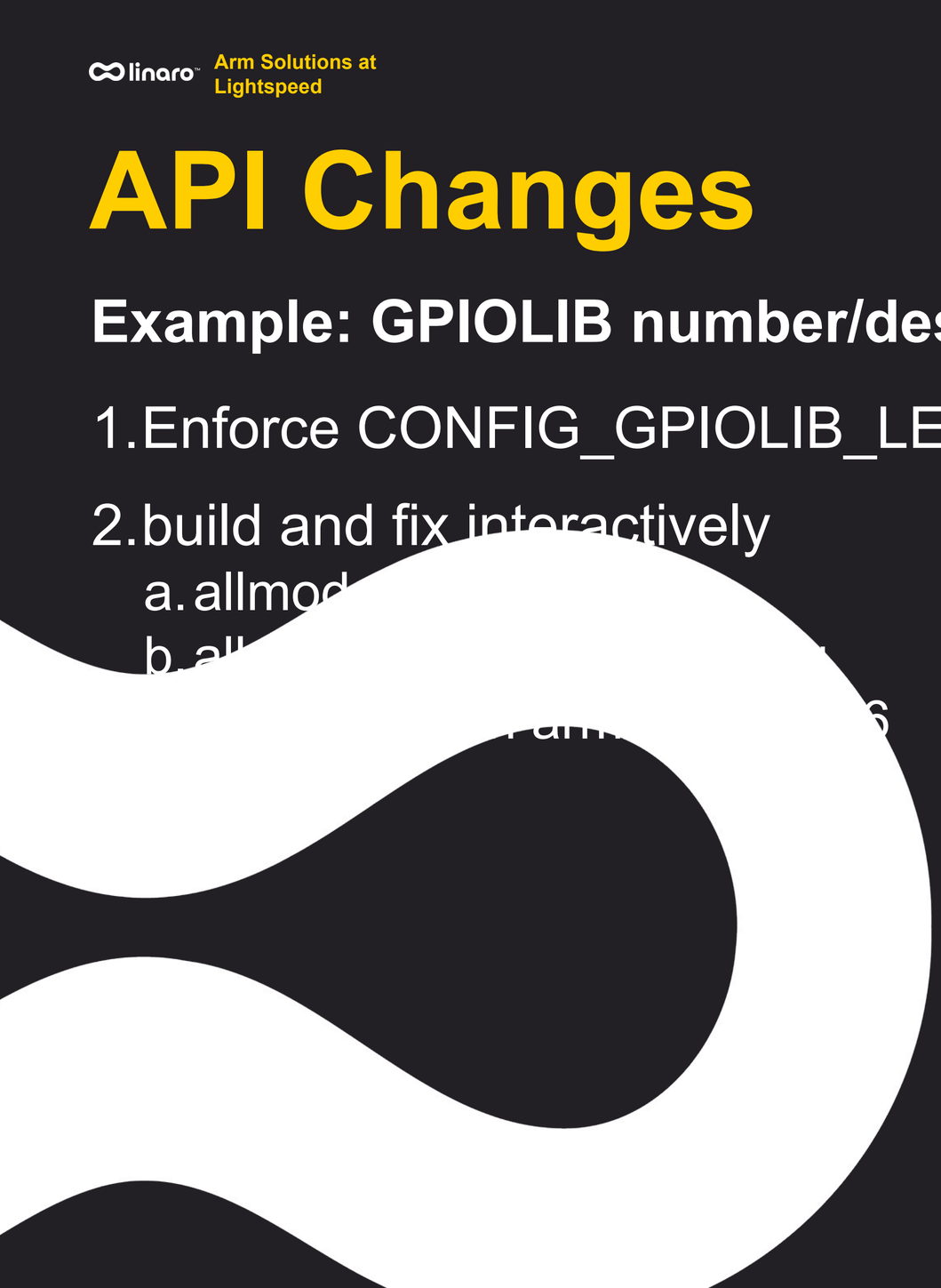


Superpowers of a build bot

API Changes

Example: GPIOLIB number/descriptor

1. Enforce CONFIG_GPIOLIB_LEGACY guard
2. build and fix interactively
   a. allmod
   b. all
   [illegible]6



Build system changes

Example: clang bringup

- Started in early 2010s
- Countless bugs in weird configurations
- Equal



Kernel warning options

Currently testing with additional options enabled

-Wframe-larger-than=1280

-Wunused-const-variable

-Wtautological-constant-out-of-range-compare

-Wformat-security

-Wformat-overflow

-Wmissing-include-

-Wmain



Future outlook

Backlog

~600 patches as of today

- many submitted before, need to recheck
- UAPI
- unds
- MA cleanups
- GFP_HIGHMEM cleanu
- emoval
- mpiler

Discussion

- Any interest in sharing the scripts?
- Anything I should test (temporarily)?
- Other

linaro.org

Thank you

Arnd Bergmann <arnd@linaro.org>